

AISSAOUI Ismail

AI & Data Science Engineer

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PROFILE

AI & Data Science Engineer specialized in LLM architectures, RAG, and cognitive support systems. Passionate about human-AI interaction and designing generative systems that extend human reflection rather than replace it. Expert in deploying Transformer models and advanced prompt engineering to support critical thinking and metacognitive engagement.

SKILLS

Programming: Python, TypeScript, JavaScript, Java, SQL

ML / DL: PyTorch, TensorFlow, Keras, Scikit-learn, Lightning, **Fine-tuning (LoRA/QLoRA)**

AI Techniques: LLMs (GPT, Llama), RAG, **Multi-Agent Systems**, **Chain-of-Thought Reasoning**

Optimization: PSO, Genetic Algorithm, ACO, Simulated Annealing, Bayesian Optimization

Web & API: React, Next.js, FastAPI, Node.js, TailwindCSS

Data & MLOps: MLflow, Airflow, Kubeflow, DVC, Pandas, NumPy

Databases: PostgreSQL, MongoDB, MySQL, Neo4j, Cassandra

Cloud & DevOps: Docker, Linux, Git, CI/CD, CUDA, AWS, Grafana

Soft Skills: Technical leadership, Communication, Stakeholder management

PROFESSIONAL EXPERIENCE

AI Engineer Intern — Digital Twin & Predictive Maintenance 2026

Sonatrach — Algeria

Developed a Digital Twin for gas turbines enabling real-time monitoring and predictive maintenance. Implemented advanced anomaly detection and RUL estimation using deep learning models.

Network Engineering Intern Sep 2024

Algérie Télécom RMC Center — Chlef, Algeria

EDUCATION

National Higher School of Computer Science (ESI-SBA) 2021–2026

Engineering and Master's Degree in Computer Science

Specialization: Artificial Intelligence & Informatics

KEY ACHIEVEMENTS

- Developed custom Transformer models (BERT, GPT, RAG) for real-world multilingual applications.
- Built explainable AI platforms for HR tech and public institutions — reducing bias and im-

proving decision quality.

- Designed a graph-based recommendation system with +24% accuracy improvement (dataset: 60K profiles).
- Delivered complete ML pipelines: data ingestion, validation, training, CI/CD, and monitoring.
- Built real-time vision systems (YOLO, ByteTrack) reducing latency by 32%.

PROJECTS

Geo-RAG — Spatial Cognitive Assistant

2025

Built a RAG system to support spatial reasoning and geospatial knowledge discovery. Focused on augmenting user analysis through interactive AI rather than automated decision-making.

Tech: ChromaDB, LangChain, OpenAI — Concepts: RAG, Cognitive Support, Spatial Logic

Aura — Enterprise AI Trading System

2025

Full-stack automated crypto trading platform powered by hybrid LSTM forecasting and **Reinforcement Learning** (Deep Q-Learning) to optimize entry/exit policies under long-term reward maximization. Includes advanced technical indicators, real-time dashboards, and interactive backtesting.

Tech: TensorFlow, Keras, FastAPI, Next.js, WebSockets — Concepts: LSTM, Reinforcement Learning, Deep Q-Learning, Time-Series Analysis

Advanced NLP Pipeline with Multi-Optimization

2025

Multilingual moderation optimized with PSO, Genetic Algorithm, and Bayesian Optimization.

Tech: DistilBERT, PyTorch, FastAPI — Concepts: Optimization, Transformers, Classification

Intelligent University Scheduler

2025

Automatic generation of conflict-free timetables using hybrid heuristics and evolutionary algorithms.

Tech: FastAPI, NumPy, Pandas — Concepts: GA, PSO, ACO, Simulated Annealing

Mini-GPT (Decoder-only)

2025

Implementation of a compact GPT model with controlled training and sequential text generation.

Tech: PyTorch, CUDA, AdamW, W&B — Concepts: Transformers, Language Modeling

North African Sentiment BERT

2025

LoRA adaptation of BERT for sentiment analysis in Maghrebi dialect.

Tech: PyTorch, LoRA, MLflow — Concepts: Fine-tuning, Low-Rank Adaptation

Geo-RAG — Map Assistant

2025

RAG system integrating geospatial data for map-oriented queries.

Tech: ChromaDB, GeoPandas, Streamlit — Concepts: RAG, Spatial Reasoning

Recognizini — Interactive AI & Feedback

2025

Facial recognition system with an active feedback loop, keeping the human in control of the learning process to preserve agency and accountability.

Tech: OpenCV, PyTorch, FastAPI — Concepts: Human-in-the-loop, Agency, Active Learning

Real-time PPE Detection (YOLOv8)

2025

Real-time detection of safety equipment (helmet, vest) for compliance monitoring.

Tech: YOLOv8, OpenCV — Concepts: Object Detection, Streaming

Traffic Analytics Suite

2025

Traffic analytics including tracking, speed estimation, and license plate recognition.

Tech: YOLOv8, ByteTrack, EasyOCR — Concepts: Tracking, OCR, Analytics

Hadj Management System

2024

End-to-end logistics platform for group management and itineraries.

Tech: React, FastAPI, PostgreSQL — Concepts: Workflow, Synchronization

Hybrid POS & E-Commerce System

2025

Offline-first POS system integrated with an online store.

Tech: PyQt, SQLite, PostgreSQL — Concepts: Offline/Online Sync, Inventory Management

LANGUAGES

Arabic: Mother Tongue | English: Advanced | French: Conversational